42. Using the method of least square, calculate the regression equation of $X$ on $Y$ and $Y$ on $X$ from the following data and estimate $X$ where $Y$ is 16.

| $x$ | 10 | 12 | 13 | 17 | 18 |
|---|---|---|---|---|---|
| $y$ | 5 | 6 | 7 | 9 | 13 |

Also determine the value of correlation coefficient.

43. The following table shows the age ($X$) and systolic blood pressure ($Y$) of 8 persons.

| Age ($X$) | 56 | 42 | 60 | 50 | 54 | 49 | 39 | 45 |
|---|---|---|---|---|---|---|---|---|
| Blood pressure ($Y$) | 160 | 130 | 125 | 135 | 145 | 115 | 140 | 120 |

Fit a simple linear regression model, $Y$ on $X$ and estimate the blood pressure of a person of 60 years.

44. Find the regression equation of $X$ on $Y$ given that $n = 5$, $\Sigma x = 30$, $\Sigma y = 40$, $\Sigma xy = 214$, $\Sigma x^2 = 220$, $\Sigma y^2 = 340$.

45. Given the following data, estimate the marks in statistics obtained by a student who has scored 60 marks in English.

Mean of marks in Statistics = 80, Mean of marks in English = 50, S.D of marks in Statistics = 15, S.D of marks in English = 10 and Coefficient of correlation = 0.4.

46. Find the linear regression equation of percentage worms ($Y$) on size of the crop ($X$) based on the following seven observations.

| Size of the crop ($X$) | 16 | 15 | 11 | 27 | 39 | 22 | 20 |
|---|---|---|---|---|---|---|---|
| Percentage worms ($Y$) | 24 | 25 | 34 | 40 | 35 | 20 | 23 |

47. In a correlation analysis, between production ($X$) and price of a commodity ($Y$) we get the following details.

Variance of $X = 36$.

The regression equations are:

$12X - 15Y + 99 = 0$ and $60X - 27Y = 321$

Calculate (a) The average value of $X$ and $Y$.

(b) Coefficient of correlation between $X$ and $Y$.



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